

WORD-COMPATIBLE GRAPHS AND GROVE POLYNOMIALS

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ABSTRACT.

1. WC GRAPHS

Definition 1.1. A *WC graph* is simply a finite subset $W \subseteq \mathbb{N} \times \mathbb{N}$ with no additional restrictions. We impose a total ordering on these pairs by declaring that $(a, b) \leq (c, d)$ if $a < c$ or $a = c$ and $b \geq d$. We define the *weight* of a WC graph W to be the sequence

$$\mathbf{wt}(W) = (i_1, \dots, i_n)$$

where i_j is the number of elements of W in row j . We define the *permutation* of a WC graph W to be the permutation

$$\mathbf{perm}(W) = s(w_1) \bullet \dots \bullet s(w_n)$$

where

$$w_i = (a_i, b_i)$$

is the i th element in the imposed ordering, and

$$s(a_i, b_i) = s_{a_i+b_i-1}$$

with $\bullet$ being the Hecke product. The *word* of W is

$$(s(w_1), \dots, s(w_n))$$

Define $\mathcal{WC}(w)$ to be the set of all WC graphs W such that $\mathbf{perm}(W) = w$.

The following is well-known.

Theorem 1.2. *We have*

$$\mathfrak{G}_w^{(\beta)}(x) = \sum_{W \in \mathcal{WC}(w)} \beta^{|W|-\ell(w)} x^{\mathbf{wt}(W)}$$

2. GROVE POLYNOMIALS AND WC GRAPHS

Definition 2.1. A labeled indexed forest (F, P) is an indexed forest F with a function (labeling) $P : \text{IN}(F) \rightarrow \mathbb{N}$ satisfying the following properties:

- (1) $P(\text{left}(u)) < P(u)$ for all $u \in \text{IN}(F)$.
- (2) $P(u) \leq P(\text{right}(u))$ for all $u \in \text{IN}(F)$.
- (3) $P(u) \geq \rho_F(u)$ for all $u \in \text{IN}(F)$.
- (4) $P(u) \leq \lambda_F(u)$ for all $u \in \text{IN}(F)$.

The set $\text{Komp}_P(F)$ is the set of functions $\kappa : \text{IN}(F) \rightarrow 2^{\mathbb{N}}$ such that

- (1) $\max(\kappa(u)) \leq P(u)$ for all $u \in \text{IN}(F)$.
- (2) $\max(\kappa(u)) \leq \min(\kappa(\text{left}(u)))$ for all $u \in \text{IN}(F)$.
- (3) $\max(\kappa(u)) < \min(\kappa(\text{right}(u)))$ for all $u \in \text{IN}(F)$.

We have that

$$\mathbf{wt}_i(\kappa) = \#\{u \in \text{in}(F) \mid i \in \kappa(u)\}$$

Theorem 2.2. Suppose a labeled indexed forest (F, P) is such that there is a $\kappa \in \text{Komp}_P(F)$ with

$$\text{wt}(\kappa) = \mathfrak{c}(F).$$

Then

$$\tilde{\mathfrak{P}}_F(x) = \sum_{\kappa \in \text{Komp}_P(F)} \beta^{|\kappa| - |F|} x^{\text{wt}(\kappa)}$$

Definition 2.3. Let $W \in \mathcal{WC}(w)$ be a WC graph for w . Define via the Nadeau-Tewari insertion algorithm

$$(P(W), Q(W)) = (P(\text{word}(W)), Q(\text{word}(W)))$$

and let

$$F(W) = \text{shape}(P(W))$$

Define $\mathfrak{c}(F)$ to be the code of the indexed forest F .

Definition 2.4. Let (F, P) be a labeled indexed forest and let $\kappa \in \text{Komp}_P(F)$. Consider the set

$$A = \bigcup_{v \in \text{IN}(F)} \kappa(v) \times \{P(v)\}$$

Define

$$\mathcal{WC}(\kappa) = \{(a_i, b_i + 1 - a_i) \mid (a_i, b_i) \in A\}$$

Theorem 2.5. For a permutation w , let $\mathcal{WC}_{\mathfrak{P}}(w, F)$ be the set of $W \in \mathcal{WC}(w)$ such that $\mathfrak{c}_{\mathfrak{P}}(W) = \mathfrak{c}(F)$. Then we have that $\mathcal{WC}(w)$ is the disjoint union of $\mathcal{WC}(\text{Komp}_{P(W)}(F(W)))$ for $W \in \mathcal{WC}_{\mathfrak{P}}(w, F)$.

Corollary 2.6. The Grothendieck polynomial $\mathfrak{G}_w^{(\beta)}(x)$ satisfies

$$\mathfrak{G}_w^{(\beta)}(x) = \sum_{F \in \text{IF}} \beta^{|F| - \ell(w)} a_w^F \tilde{\mathfrak{P}}_F(x)$$

where

$$a_w^F = \#\{W \in \mathcal{WC}_{\mathfrak{P}}(w, F) \mid \text{wt}(W) = \mathfrak{c}(F)\}$$

Theorem 2.7. Let a be a composition. Let w be the permutation such that $\mathfrak{c}(w) = a$. Then we have that

$$\tilde{\mathfrak{P}}_a(x) = \sum_{W \in \mathcal{WC}_{\mathfrak{P}}(w, a) \cap \mathcal{WC}_{\text{QY}}(w)} \mathcal{G}_{\text{wt}(W)}(x)$$

Definition 2.8. We define an invariant $\tilde{P}_{\mathfrak{P}}(W)$ of a WC graph W to be the WC graph C such that $\mathfrak{c}_{\mathfrak{P}}(C) = \text{wt}(C) = \mathfrak{c}_{\mathfrak{P}}(W)$ and there exists a $\kappa \in \text{Komp}_{P(C)}(F(C))$ such that $W \in \mathcal{WC}(\kappa)$.

Theorem 2.9 (Dual Grove LR rule). Suppose

$$\tilde{\mathfrak{P}}_a^* \sqcup \tilde{\mathfrak{P}}_b^* = \sum_c P_{ab}^c \tilde{\mathfrak{P}}_c^*$$

for integers P_{ab}^c , with $\ell(a) = p$ and $\ell(b) = q$. Let C be a WC graph with $\text{ht}(C) = p + q$ such that $\mathfrak{c}_{\mathfrak{P}}(C) = \text{wt}(C) = c$. Then P_{ab}^c is the number of WC graphs W with $\tilde{P}_{\mathfrak{P}}(W) = \tilde{P}_{\mathfrak{P}}(C)$ such that $\mathfrak{c}_{\mathfrak{P}}(\text{clip}^p(W)) = \text{wt}(\text{clip}^p(W)) = a$ and $\mathfrak{c}_{\mathfrak{P}}(\text{trim}^p(W)) = \text{wt}(\text{trim}^p(W)) = b$.

3. HECKE INSERTION, K -KNUTH EQUIVALENCE, AND LASCoux CLASSES

The purpose of this section is to attach to every WC graph a *Lascoux class*, namely the K -Knuth equivalence class of the increasing tableau obtained by Hecke insertion of its word. The main input is a theorem, due to Reiner–Yong [5] (conjecture) and Shimozono–Yu [7] (proof), stating that the WC graphs of w lying in a single K -Knuth class have generating function equal to a single Lascoux polynomial. This refines the symmetric statement of Buch–Kresch–Shimozono–Tamvakis–Yong [1]. We phrase everything in the WC-graph language of the previous sections, so that it can be used to show that WC-graph transition, and hence the squash product, preserves Lascoux classes.

Theorem 3.1 ([1, 2]). For words c, c' of positive integers the following hold.

- (1) $P(c) = P(c')$ if and only if $c \stackrel{K}{\equiv} c'$; thus the insertion tableau is a complete invariant of the K -Knuth class.

- (2) The Hecke product $s_{c_1} \bullet \cdots \bullet s_{c_n}$ depends only on the K -Knuth class of c ; it equals the Hecke product of the row reading word of $P(c)$.

Let ∂_i denote the i th divided difference operator,

$$\partial_i f = \frac{f - s_i f}{x_i - x_{i+1}},$$

and define the K -theoretic isobaric (Demazure) operator

$$\pi_i^{(\beta)} f = \partial_i (x_i (1 + \beta x_{i+1}) f).$$

Then $\pi_i^{(\beta)}(1) = 1$, $\pi_i^{(\beta)}$ sends polynomials to polynomials whose value is symmetric in x_i, x_{i+1} , and at $\beta = 0$ it recovers the Demazure operator $\pi_i f = \partial_i(x_i f)$ of the key polynomials.

Definition 3.2. For a weak composition $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$ the Lascoux polynomial $\mathfrak{L}_\alpha$ is defined by:

- (1) if $\alpha_1 \geq \alpha_2 \geq \cdots \geq \alpha_n$ (i.e. α is a partition), then $\mathfrak{L}_\alpha = x^\alpha = \prod_i x_i^{\alpha_i}$;
- (2) if $\alpha_i < \alpha_{i+1}$ for some i , then $\mathfrak{L}_\alpha = \pi_i^{(\beta)} \mathfrak{L}_{s_i \alpha}$, where $s_i \alpha$ interchanges α_i and α_{i+1} .

This is well defined by [7]. The lowest- β component of $\mathfrak{L}_\alpha$ (its terms of total degree $|\alpha|$, i.e. the β^0 part) is the key (Demazure) polynomial κ_α .

3.1. The Grothendieck-to-Lascoux expansion. Recall from the definition of a WC graph that the elements of W are listed as $w_1 < w_2 < \cdots < w_n$ in the total order on $\mathbb{N} \times \mathbb{N}$, with $w_i = (a_i, b_i)$, and that

$$s(a_i, b_i) = s_{a_i+b_i-1}.$$

Definition 3.3. For a WC graph W , define the *diagonal skew tableau* $\mathbf{dw}(W)$ to be the unique tableau of skew shape $\text{st}(|W|)/\text{st}(|W| - 1)$ whose reading word is $\text{word}(W)$. The Lascoux invariant $\mathfrak{L}(W)$ of W is the greedy rectification of $\mathbf{dw}(W)$. We can in fact apply this to any word.

Theorem 3.4 (Dual Lascoux LR rule). *Suppose*

$$\mathcal{L}_a^* \sqcup \mathcal{L}_b^* = \sum_c L_{ab}^c \mathcal{L}_c^*$$

for integers L_{ab}^c , with $\ell(a) = p$ and $\ell(b) = q$. Let C be a WC graph with $\mathbf{ht}(C) = p + q$ such that $\mathbf{wt}(\mathfrak{L}(C)) = \mathbf{wt}(C) = c$. Then L_{ab}^c is the number of WC graphs W with $\mathfrak{L}(W) = \mathfrak{L}(C)$ such that $\mathbf{wt}(\mathfrak{L}(\text{clip}^p(W))) = \mathbf{wt}(\text{clip}^p(W)) = a$ and $\mathbf{wt}(\mathfrak{L}(\text{trim}^p(W))) = \mathbf{wt}(\text{trim}^p(W)) = b$.

The next theorem is the symmetric statement of Buch–Kresch–Shimozono–Tamvakis–Yong, in the WC-graph normalization: grouping the WC graphs of w by Lascoux class collects their generating function into symmetric Grothendieck building blocks.

Theorem 3.5 ([1]). *Fix an increasing tableau P of straight shape λ whose row reading word has Hecke product w . Then*

$$\sum_{\substack{W \in \mathcal{WC}(w) \\ P(W) = P}} \beta^{|W| - \ell(w)} x^{\mathbf{wt}(W)}$$

is symmetric in the variables it involves; summed over all such P of a fixed shape λ it equals the stable (Grassmannian) Grothendieck contribution $a_{w, \lambda} G_\lambda$, recovering the factor-sequence expansion $G_w = \sum_\lambda a_{w, \lambda} G_\lambda$.

The nonsymmetric refinement replaces each symmetric block by a single Lascoux polynomial. This is the “Grothendieck to Lascoux” expansion.

Theorem 3.6 (Reiner–Yong [5]; Shimozono–Yu [7]). *There is a map $P \mapsto \alpha(P)$ assigning to each increasing tableau P a weak composition $\alpha(P)$, with underlying partition $\text{sh}(P)$, determined by the K -theoretic right key of P , such that for every increasing tableau P arising as an insertion tableau of some $W \in \mathcal{WC}(w)$,*

$$\sum_{\substack{W \in \mathcal{WC}(w) \\ P(W) = P}} \beta^{|W| - \ell(w)} x^{\mathbf{wt}(W)} = \mathfrak{L}_{\alpha(P)}.$$

Consequently

$$\mathfrak{G}_w^{(\beta)}(x) = \sum_{P \in \mathcal{I}(w)} \mathfrak{L}_{\alpha(P)} = \sum_{\alpha} c_{w,\alpha} \mathfrak{L}_{\alpha}, \quad c_{w,\alpha} = \#\{P \in \mathcal{I}(w) \mid \alpha(P) = \alpha\} \in \mathbb{Z}_{\geq 0},$$

where $\mathcal{I}(w) = \{P(W) \mid W \in \mathcal{WC}(w)\}$ is the set of Lascoux classes of w .

The content we use below is the following: the Lascoux polynomial attached to a WC graph depends only on its Lascoux class. We isolate this as a corollary.

Corollary 3.7. *The assignment*

$$W \longmapsto \mathfrak{L}_{\alpha(P(W))}$$

factors through the Lascoux class $[W] = P(W)$. In particular, if Φ is any map on WC graphs with $P(\Phi(W)) \stackrel{K}{\equiv} P(W)$ for all W , equivalently $\mathbf{dw}(\Phi(W)) \stackrel{K}{\equiv} \mathbf{dw}(W)$, then

$$\mathfrak{L}_{\alpha(P(\Phi(W)))} = \mathfrak{L}_{\alpha(P(W))}$$

for all W ; that is, Φ preserves Lascoux classes and hence the associated Lascoux polynomial.

Proof. Immediate from Theorem 3.1(1), which identifies $P(W)$ with the K -Knuth class of $\mathbf{dw}(W)$, and Theorem 3.6, in which α is a function of $P(W)$ alone. $\square$

Remark. Corollary 3.7 is the K -theoretic, straight-shape analogue of the grove/forest decomposition of Section 3's predecessor: there the WC graphs of w are grouped by the indexed forest $F(W)$ of the Nadeau–Tewari insertion and each fiber contributes $\mathfrak{P}_{F(W)}$, whereas here they are grouped by the K -Knuth class $P(W)$ of Hecke insertion and each fiber contributes $\mathfrak{L}_{\alpha(P(W))}$. Identifying the two groupings, e.g. establishing an equality $\tilde{\mathfrak{P}}_F = \mathfrak{L}_{\alpha(P)}$ under the correspondence $F \leftrightarrow P$, would identify grove polynomials with Lascoux polynomials.

4. BOUNDED WC GRAPHS AND THE K -THEORETIC TRANSITION

We now equip WC graphs with a row grading and a transition map, mirroring the treatment of bounded RC graphs in the reduced theory of [6]. Everything in this section specializes to those reduced constructions on the subset of RC graphs.

Definition 4.1. A *bounded WC graph* is a WC graph W together with a specified number of rows $n \geq \max\{a : (a, b) \in W \text{ for some } b\}$ subject to $\mathbf{m}(\text{perm}(W)) \leq n$. We write (W, n) for the pair, abuse notation to treat the number of rows as a property of W , and call $\text{ht}(W) = n$ the *height* of W . For $\text{ht}(W) \geq 1$ we define the *trim* $\text{trim}(W)$ to be the bounded WC graph with height $\text{ht}(W) - 1$ and underlying set

$$\text{trim}(W) = \{(a-1, b-1) : (a, b) \in W, a \geq 2\},$$

and we set $\uparrow^m W = \{(a+m, b) \mid (a, b) \in W\}$; these are the bounded RC-graph operations of [6] applied to WC graphs verbatim. Let $\mathcal{BWC}$ be the free abelian group on all bounded WC graphs, graded by height,

$$\mathcal{BWC} = \bigoplus_{n=0}^{\infty} \mathcal{BWC}_n.$$

A WC graph with $\ell(\text{perm}(W)) = |W|$ is exactly an RC graph, and on such graphs trim and $\uparrow^m$ agree with the RC operations. Thus $\mathcal{BRC} \subseteq \mathcal{BWC}$ is a graded subgroup and the projection $\rho : \mathcal{BWC} \rightarrow \mathcal{BRC}$ that kills every WC graph with $|W| > \ell(\text{perm}(W))$ is a graded retraction.

4.1. The transition map via marked bumpless pipe dreams. In the reduced case the map $\mathcal{Z}$ of [6] realizes the Lascoux–Schützenberger transition on RC graphs. Its K -theoretic counterpart is most naturally described through bumpless pipe dreams. Recall the following ingredients.

- (1) **Marked bumpless pipe dreams.** Weigandt [8] introduced *marked bumpless pipe dreams* (marked BPDs) and proved that the double Grothendieck polynomial $\mathfrak{G}_w^{(\beta)}$ is the generating function of marked BPDs of w , extending the Lam–Lee–Shimozono formula from K -theory. Let $\text{MBPD}(w)$ denote the set of marked BPDs of w .

- (2) **The HSY bijection.** Huang, Shimozono, and Yu [4] construct a weight-preserving bijection between marked BPDs and reverse compatible pairs, the latter being in bijection with not-necessarily-reduced pipe dreams; their bijection generalizes, in the unmarked reduced case, the canonical bijection of Gao and Huang [3]. Reading a bounded WC graph as the compatible sequence of a (not necessarily reduced) pipe dream, this yields a weight-preserving bijection

$$\Psi : \bigcup_n \{W \in \mathcal{BWC}_n : \text{perm}(W) = w\} \longrightarrow \text{MBPD}(w),$$

whose restriction to RC graphs is the Gao–Huang bijection ϕ used in the reduced theory of [6].

- (3) **Weigandt transition.** Weigandt’s droop/transition operation $\mathbf{t}$ on bumpless pipe dreams removes the bottom row and rearranges the resulting diagram into a valid BPD of one fewer row, realizing the (co)transition recursion for Grothendieck polynomials; it acts on marked BPDs with an empty bottom row [8, 4].

Definition 4.2. Let W be a bounded WC graph with $\text{ht}(W) = n$ whose row n is empty. Define the K -theoretic transition map

$$\mathcal{Z}(W) = \Psi^{-1}(\mathbf{t}(\Psi(W))),$$

a bounded WC graph of height $n - 1$. We extend $\mathcal{Z}$ to an endomorphism of $\mathcal{BWC}$ by setting $\mathcal{Z}(W) = 0$ when row n is nonempty.

The following properties are the K -theoretic analogues of the reduced-case transition lemma, the trim-commutation lemma, and the zeroing bijection of [6]. We record them as the working hypotheses of the construction; each specializes to its reduced counterpart under ρ , where it is a theorem of [6].

Proposition 4.3. *On RC graphs the map $\mathcal{Z}$ agrees with $\mathcal{Z}$: for every bounded WC graph W with empty last row,*

$$\rho(\mathcal{Z}(W)) = \mathcal{Z}(\rho(W)),$$

and in particular $\mathcal{Z}|_{\mathcal{BRC}} = \mathcal{Z}$.

Proof sketch. On RC graphs Ψ is the Gao–Huang bijection ϕ and $\mathbf{t}$ restricts to Weigandt’s transition on reduced BPDs, so $\mathcal{Z} = \phi^{-1} \circ \mathbf{t} \circ \phi$ there. The reduced-case identity $\mathcal{Z} \circ \phi^{-1} = \phi^{-1} \circ \mathbf{t}$ (the conjecture of [6] relating $\mathcal{Z}$ and $\mathbf{t}$ through ϕ) gives the claim; ρ discards the non-reduced terms, none of which are produced from a reduced input. $\square$

Proposition 4.4. *Let W be a bounded WC graph with $\text{ht}(W) = n \geq 1$ and empty row n . Then $\text{ht}(\mathcal{Z}(W)) = n - 1$, the weight is preserved,*

$$\text{wt}(\mathcal{Z}(W)) = \text{wt}(W),$$

and the permutation undergoes the K -theoretic (Grothendieck) transition: $\text{perm}(\mathcal{Z}(W))$ is obtained from $\text{perm}(W)$ by a single step of the Grothendieck cotransition recursion in row n .

Proposition 4.5. *Let (W, n) be a bounded WC graph with $n \geq 2$ whose row n is empty. Then*

$$\mathcal{Z}(\text{trim}(W)) = \text{trim}(\mathcal{Z}(W)).$$

Proof sketch. This is the K -analogue of the trim-commutation lemma of [6]. Trimming preserves a suffix of the word of W , and both Ψ and $\mathbf{t}$ act on the bottom rows of the marked BPD without reference to the top row: the transition $\mathbf{t}$ proceeds downward from the removed bottom row, so discarding the top row commutes with it. Under ρ the identity specializes to the reduced statement of [6]. $\square$

Proposition 4.6. *Let w be a permutation with last descent at most n , and let $\mathcal{WC}(w, n)^0$ be the set of bounded WC graphs W with $\text{ht}(W) = n$, $\text{perm}(W) = w$, and empty row n . Then*

$$\mathcal{Z} : \mathcal{WC}(w, n)^0 \longrightarrow \bigcup_{w \xrightarrow{n} w'} \mathcal{WC}(w', n - 1)$$

is a weight-preserving bijection, where w' ranges over the outputs of the Grothendieck cotransition in row n . Consequently

$$\mathcal{Z}\left(\sum_{\substack{\text{perm}(W)=w \\ \text{ht}(W)=n}} W\right) = \sum_{w \xrightarrow{n} w'} \sum_{\substack{\text{perm}(W')=w' \\ \text{ht}(W')=n-1}} W',$$

the WC-graph form of the Grothendieck transition formula.

5. WC GRAPH RING

We now lift the K -theoretic transition map $\mathcal{Z}$ to an associative product on $\mathcal{BWC}$, in exact parallel with the product $\sqcup$ on $\mathcal{BRC}$ built from $\mathcal{Z}$ in [6]. Only the underlying zeroing map has changed: $\mathbf{trim}^p$ is unchanged, and $\mathbf{clip}^p$ is now built from $\mathcal{Z}$.

Definition 5.1. Let W be a bounded WC graph with $\mathbf{ht}(W) = n$ and let $0 \leq p \leq n$. Define $\mathbf{trim}^p(W) \in \mathcal{BWC}_{n-p}$ to be the result of discarding the top p rows of W and reindexing (equivalently, the p -fold iterate of $\mathbf{trim}$), and define $\mathbf{clip}^p(W) \in \mathcal{BWC}_p$ to be the $(n-p)$ -fold iterate of $\mathcal{Z}$ applied to the bottom row,

$$\mathbf{clip}^p(W) = \mathcal{Z}^{n-p}(W),$$

which is defined whenever each intermediate bottom row is empty at the moment it is zeroed (and is 0 otherwise). Thus $\mathbf{clip}^p(W)$ and $\mathbf{trim}^p(W)$ decompose W across the horizontal cut between rows p and $p+1$.

Associativity of the product below rests on two structural identities relating $\mathbf{clip}^p$ and $\mathbf{trim}^p$: functoriality of $\mathbf{clip}$ under iteration, and commutativity of $\mathbf{clip}$ with $\mathbf{trim}$. The first is formal; the second is the content of Proposition 4.5.

Lemma 5.2. Let W be a bounded WC graph with $\mathbf{ht}(W) = n$ and $0 \leq p \leq p+q \leq n$. Then

$$\mathbf{clip}^p(\mathbf{clip}^{p+q}(W)) = \mathbf{clip}^p(W).$$

Proof. Both sides are computed by iterating $\mathcal{Z}$ on the bottom row. With $r = n - (p+q)$, the graph $\mathbf{clip}^{p+q}(W)$ is the r -fold iterate $\mathcal{Z}^r(W)$ and $\mathbf{clip}^p(W) = \mathcal{Z}^{r+q}(W)$; the first r iterations in the computation of $\mathbf{clip}^p(W)$ produce $\mathbf{clip}^{p+q}(W)$, and the remaining q produce $\mathbf{clip}^p(\mathbf{clip}^{p+q}(W))$. Hence the two agree. $\square$

Lemma 5.3 (Commutativity of $\mathbf{clip}$ and $\mathbf{trim}$). Let W be a bounded WC graph with $\mathbf{ht}(W) = p+q+r$. Then

$$\mathbf{trim}^p(\mathbf{clip}^{p+q}(W)) = \mathbf{clip}^q(\mathbf{trim}^p(W)).$$

Proof. Both sides have height q . By Proposition 4.5, $\mathcal{Z}(\mathbf{trim}(W')) = \mathbf{trim}(\mathcal{Z}(W'))$ whenever the last row of W' is empty. Since $\mathbf{clip}^{p+q}$ on a graph of height $p+q+r$ and $\mathbf{clip}^q$ on a graph of height $q+r$ are both the r -fold iterate of $\mathcal{Z}$ on the bottom row (cf. Lemma 5.2), iterating Proposition 4.5 r times gives

$$\mathbf{trim}^p(\mathbf{clip}^{p+q}(W)) = \mathbf{trim}^p(\mathcal{Z}^r(W)) = \mathcal{Z}^r(\mathbf{trim}^p(W)) = \mathbf{clip}^q(\mathbf{trim}^p(W)),$$

as required. $\square$

Definition 5.4. For bounded WC graphs W_1 with $\mathbf{ht}(W_1) = m$ and W_2 with $\mathbf{ht}(W_2) = n$, define

$$W_1 \sqcup W_2 = \sum_{\substack{\mathbf{clip}^m(W')=W_1 \\ \mathbf{trim}^n(W')=W_2}} W',$$

where W' ranges over bounded WC graphs with $\mathbf{ht}(W') = m+n$, and extend bilinearly to $\mathcal{BWC}$.

Theorem 5.5. The product $\sqcup$ is associative, and $\mathcal{BWC}$ is a ring under $\sqcup$ with identity the empty WC graph of height 0.

Proof. Let W_1, W_2, W_3 be bounded WC graphs of heights p, q, r , and let W be a bounded WC graph with $\mathbf{ht}(W) = p+q+r$. Unwinding Definition 5.4 in each bracketing, W contributes to the doubly-bracketed products under the conditions

	Left $(W_1 \sqcup W_2) \sqcup W_3$	Right $W_1 \sqcup (W_2 \sqcup W_3)$
(i)	$\mathbf{clip}^p(\mathbf{clip}^{p+q}(W)) = W_1$	$\mathbf{clip}^p(W) = W_1$
(ii)	$\mathbf{trim}^p(\mathbf{clip}^{p+q}(W)) = W_2$	$\mathbf{clip}^q(\mathbf{trim}^p(W)) = W_2$
(iii)	$\mathbf{trim}^{p+q}(W) = W_3$	$\mathbf{trim}^q(\mathbf{trim}^p(W)) = W_3$

Row (i) is equivalent on both sides by Lemma 5.2, row (ii) by Lemma 5.3, and row (iii) because $\mathbf{trim}^{p+q} = \mathbf{trim}^q \circ \mathbf{trim}^p$ by definition of $\mathbf{trim}$. Each condition is an equality carrying no combinatorial multiplicity, so W occurs with multiplicity 1 in either bracketing, and the two bracketed products agree. Distributivity is imposed by bilinear extension, and the empty height-0 WC graph is a two-sided identity. $\square$

Because $\mathcal{Z}$ restricts to $\mathcal{Z}$ on RC graphs (Proposition 4.3) and $\mathbf{trim}$ is unchanged, the retraction ρ intertwines the two clip/trim decompositions; hence it is compatible with the products.

Proposition 5.6. *The linear map*

$$\rho : \mathcal{BWC} \rightarrow \mathcal{BRC}$$

such that

$$\rho(W) = \begin{cases} W & \text{if } W \text{ is an RC graph} \\ 0 & \text{otherwise} \end{cases}$$

is a surjective ring homomorphism.

6. THE K -PLACTIC MONOID AND ALGEBRA

The reduced theory of [6] attaches to each rank k a *plactic monoid* Plac_k : the set of height- k RC graphs whose permutation is k -Grassmannian, made into a monoid by the squash product, which is a combinatorial model for the ring of Schur polynomials in k variables. We now introduce its K -theoretic, set-valued analogue. It is obtained by replacing RC graphs with WC graphs and, correspondingly, semistandard Young tableaux and Schur polynomials with their set-valued and stable Grothendieck counterparts.

Recall that a permutation w is *k -Grassmannian* if it is either the identity or has a unique descent, located at position k . For such a w the associated partition

$$\lambda(w) = (w(k) - k, w(k-1) - (k-1), \dots, w(1) - 1)$$

has at most k parts, and the Grothendieck polynomial $\mathfrak{G}_w^{(\beta)}$, expressed in the variables $x_1, \dots, x_k$, is the stable (Grassmannian) Grothendieck polynomial $G_{\lambda(w)}(x_1, \dots, x_k)$; its lowest-degree component is the Schur polynomial $s_{\lambda(w)}(x_1, \dots, x_k)$. By the WC-graph formula $\mathfrak{G}_w^{(\beta)}(x) = \sum_{W \in \mathcal{WC}(w)} \beta^{|W| - \ell(w)} x^{\mathbf{wt}(W)}$, the height- k WC graphs of a k -Grassmannian permutation are the set-valued analogue of the semistandard tableaux enumerated by a Schur polynomial.

Definition 6.1 (Squash product). Let W_1 and W_2 be bounded WC graphs of height k . Their *disjoint union* $W_1 \star W_2$ is formed by shifting every entry of W_2 upward far enough that each reflection $s(a, b)$ used by W_2 exceeds—and hence Hecke commutes with—every reflection used by W_1 , and then overlaying the two graphs row by row. The weight is additive, $\mathbf{wt}(W_1 \star W_2) = \mathbf{wt}(W_1) + \mathbf{wt}(W_2)$. The *squash product* is

$$W_1 \boxtimes W_2 = \text{clip}^k(W_1 \star W_2),$$

the clip of the disjoint union back down to k rows (Definition 5.1).

On k -Grassmannian factors the squash product is exactly the single surviving term of the ring product $\boxplus$ of Definition 5.4: since the clip clip^k and trim $\mathbf{trim}^k$ of $W_1 \star W_2$ recover W_1 and W_2 respectively,

$$W_1 \boxtimes W_2 = W_1 \boxplus W_2 \quad (\mathbf{ht}(W_1) = \mathbf{ht}(W_2) = k).$$

In particular $\boxtimes$ is associative, being the restriction of the associative product $\boxplus$ (Theorem 5.5) to n -Grassmannian WC graphs.

Definition 6.2 (K -plactic monoid and algebra). The *K -plactic monoid of rank k* is the set

$$\text{Plac}_k^K = \{ W \in \mathcal{BWC}_k : \mathbf{perm}(W) \text{ is } k\text{-Grassmannian} \}$$

of height- k WC graphs whose permutation has no descent or a single descent at k , equipped with the squash product $\boxtimes$. It is a monoid with unit the empty height- k WC graph. The *K -plactic algebra* $\mathbb{Z}\text{Plac}_k^K$ is its monoid algebra: the free abelian group on Plac_k^K with the bilinear extension of $\boxtimes$.

Theorem 6.3. *The set Plac_k^K is closed under $\boxtimes$, and $(\text{Plac}_k^K, \boxtimes)$ is an associative monoid. It is the set-valued analogue of the plactic monoid Plac_k of semistandard tableaux of at most k rows: the degeneration that discards every WC graph W with $|W| > \ell(\mathbf{perm}(W))$ (equivalently, restricts to RC graphs) sends Plac_k^K onto Plac_k and $\boxtimes$ to the classical squash product.*

Proof. Closure holds because clip^k carries a height- $2k$ WC graph whose top and bottom halves are k -Grassmannian to a height- k WC graph whose permutation is again k -Grassmannian: the transition $\mathcal{Z}$ (Proposition 4.4) preserves the property of having a single descent at the top row. Associativity is the restriction of Theorem 5.5. The degeneration is the restriction of the retraction $\rho : \mathcal{BWC} \rightarrow \mathcal{BRC}$ (which is a ring homomorphism), and on RC graphs $\text{clip}^k = \mathcal{Z}^k$ agrees with the reduced clip of [6], so ρ intertwines $\boxtimes$ with the reduced squash product. $\square$

The squash product realizes stable Grothendieck multiplication, in exact parallel with the crystal isomorphism of the reduced theory [6]. We record this as the K -theoretic analogue.

Theorem 6.4. *Let u, v be k -Grassmannian permutations. Then the disjoint union induces a weight-preserving bijection*

$$\{W \in \text{Plac}_k^K : \text{perm}(W) = u\} \times \{W' \in \text{Plac}_k^K : \text{perm}(W') = v\} \longrightarrow \{U \in \mathcal{BWC}_{2k} : \text{clip}^k(U) \in \text{Plac}_k^K(u), \text{trim}^k(U) \in \text{Plac}_k^K(v)\}$$

and clipping back to k rows decomposes the result according to the K -theoretic Littlewood–Richardson (Buch) rule [1]: writing λ, μ for the partitions of u, v ,

$$\sum_{\substack{W \in \text{Plac}_k^K(u) \\ W' \in \text{Plac}_k^K(v)}} W \boxtimes W' = \sum_{\nu} c_{\lambda\mu}^{\nu}(\beta) \sum_{\substack{W'' \in \text{Plac}_k^K \\ \lambda(\text{perm}(W'')) = \nu}} W'',$$

where $c_{\lambda\mu}^{\nu}(\beta)$ are the structure constants of the stable Grothendieck polynomials, $G_{\lambda}G_{\mu} = \sum_{\nu} c_{\lambda\mu}^{\nu}(\beta)G_{\nu}$.

For a k -Grassmannian permutation w we assemble the WC graphs of w into a single element of the K -plactic algebra.

Definition 6.5. For a k -Grassmannian permutation w , the *Grothendieck element* is

$$\mathbf{G}_w = \sum_{\substack{W \in \text{Plac}_k^K \\ \text{perm}(W) = w}} W \in \mathbb{Z}\text{Plac}_k^K,$$

the sum of all height- k WC graphs with permutation w . Under the weight map $W \mapsto \beta^{|W|-\ell(w)}x^{\text{wt}(W)}$ it maps to the stable Grothendieck polynomial $G_{\lambda(w)}(x_1, \dots, x_k)$.

Theorem 6.6. *As w ranges over the k -Grassmannian permutations, the Grothendieck elements $\mathbf{G}_w$ are linearly independent and span a commutative subring $\mathbf{G}_k \subseteq \mathbb{Z}\text{Plac}_k^K$. The weight map induces a ring isomorphism*

$$\mathbf{G}_k \xrightarrow{\sim} \mathbb{Z}[G_{\lambda}(x_1, \dots, x_k) : \lambda \text{ a partition with } \leq k \text{ parts}], \quad \mathbf{G}_w \mapsto G_{\lambda(w)}(x_1, \dots, x_k),$$

onto the ring generated by the stable Grothendieck polynomials in k variables. Passing to lowest degree recovers the classical model: $\mathbf{G}_k$ degenerates to the ring of symmetric polynomials spanned by the Schur polynomials $s_{\lambda}(x_1, \dots, x_k)$, and $\mathbf{G}_w$ to the Schur element of Plac_k .

Proof. Distinct k -Grassmannian permutations have disjoint sets of WC graphs, so the $\mathbf{G}_w$ are linearly independent. By Theorem 6.4 the squash product of two Grothendieck elements decomposes with the same structure constants as the product of the corresponding stable Grothendieck polynomials,

$$\mathbf{G}_u \boxtimes \mathbf{G}_v = \sum_w c_{u,v}^w(\beta) \mathbf{G}_w, \quad G_{\lambda(u)} G_{\lambda(v)} = \sum_w c_{u,v}^w(\beta) G_{\lambda(w)},$$

where $c_{u,v}^w(\beta)$ are the K -theoretic Littlewood–Richardson (Buch) coefficients. Hence the span $\mathbf{G}_k$ is a subring, the assignment $\mathbf{G}_w \mapsto G_{\lambda(w)}$ is a ring homomorphism, and it is bijective onto the stated ring because the stable Grothendieck polynomials of partitions with at most k parts are a $\mathbb{Z}$ -basis of it. Commutativity follows since the target ring is commutative. Setting $\beta = 0$ (equivalently, retaining the lowest-degree part, which is ρ followed by symmetrization) sends $G_{\lambda} \mapsto s_{\lambda}$ and recovers the plactic model of [6]. $\square$

Remark. Theorem 6.6 is the K -theoretic analogue of the statement that, inside the plactic algebra, the sums of tableaux of fixed shape span a commutative ring isomorphic to the ring of Schur polynomials. Here shapes are replaced by k -Grassmannian permutations, tableaux by WC graphs, Knuth equivalence by K -Knuth equivalence (Section 3), and Schur polynomials by stable Grothendieck polynomials.

The K -plactic algebras also carry the degeneration to the reduced plactic algebras and assemble into a tensor model for $\mathcal{BWC}_n$.

Definition 6.7. For each $n > 0$ there is a surjective ring homomorphism

$$\mathbb{Z}\text{Plac}_n^K \rightarrow \mathbb{Z}\text{Plac}_n$$

obtained on elementary symmetric WC graphs by sending $E_{p,k}^\beta$ with $|\beta| > p$ to $E_{|\beta|,k}^\beta$; it is the restriction of the retraction ρ and provides a section of the inclusion $\text{Plac}_n \hookrightarrow \text{Plac}_n^K$. Define

$$\mathcal{M}_n^K = \bigotimes_{i=1}^n \mathbb{Z}\text{Plac}_i^K.$$

There is an evident homomorphism $\mathcal{M}_n^K \rightarrow \mathcal{M}_n$ obtained by applying the above homomorphism to each factor, giving a commutative diagram

$$\begin{array}{ccc} \mathcal{M}_n^K & \longrightarrow & \mathcal{M}_n \\ \downarrow & & \downarrow \\ \mathcal{BWC}_n & \dashrightarrow & \mathcal{BRC}_n \end{array}$$

Remark. The compatibility of $\boxplus$ with the Lascoux decomposition of Theorem 3.6 is deferred. Corollary 3.7 shows that any map preserving the K -Knuth class $\mathbf{P}(W)$ preserves the associated Lascoux polynomial; the outstanding question is how $\mathcal{Z}$, and hence clip^p and the squash product, interact with the classes $\mathbf{P}(W)$. We return to this once the properties of $\mathcal{Z}$ in Propositions 4.4–4.6 are established.

7. FORMULAS

Define the double grothendieck polynomial $\mathfrak{G}_w(x; y)$ by

$$\mathfrak{G}_w^{(\beta)}(x; y) = \sum_{W \in \text{WC}(w)} \beta^{|W| - \ell(w)} (x \ominus y)^W$$

where

$$(x \ominus y)^W = \prod_{(i,j) \in W} \left(\frac{x_i - y_j}{1 + \beta y_j} \right)$$

$$\omega^i = \frac{(1 + \beta t_i) - s_i(1 + \beta t_i)}{t_i - t_{i+1}} = \frac{(1 - s_i)(1 + \beta t_i)}{t_i - t_{i+1}} = \delta^i(1 + \beta t_i)$$

where δ^i is the divided difference operator on the t variables. Leibniz rule,

$$\omega^i(fg) = \omega^i(f)g + f(\delta^i + s_i)g$$

Let $\mathbf{w}_1 \cdots \mathbf{w}_n$ be a reduced word for w . Let $\mathbf{u}_1 \cdots \mathbf{u}_n$ be a subword, meaning $\mathbf{u}_i \in \{1, \mathbf{w}_i\}$. Let $\mathbf{v}$ be a second subword in the complementary spaces of $\mathbf{u}$, meaning $\mathbf{v}_i \in \{1, \mathbf{w}_i\}$ and $\mathbf{v}_i = 1$ if $\mathbf{u}_i = \mathbf{w}_i$. Then let

$$\omega^{\mathbf{u}} = \omega^{\mathbf{u}_1} \cdots \omega^{\mathbf{u}_n}$$

and let

$$\pi^i = \begin{cases} \omega^i & \text{if } \mathbf{v}_i = \mathbf{w}_i \\ 1 & \text{if } \mathbf{u}_i = \mathbf{w}_i \\ \delta^i + s_i & \text{if } \mathbf{u}_i = 1 \text{ and } \mathbf{v}_i = 1 \end{cases}$$

and let

$$\omega_{\mathbf{v}}^{\mathbf{w}/\mathbf{u}} = \pi^1 \cdots \pi^n$$

Then we have the following general Leibniz rule:

$$\omega^w(fg) = \sum_{\mathbf{u}, \mathbf{v}} \omega^{\mathbf{u}}(f) \omega_{\mathbf{v}}^{\mathbf{w}/\mathbf{u}}(g)$$

where the sum is over all subwords $\mathbf{u}$ and $\mathbf{v}$ in complementary spaces of $\mathbf{u}$.

To make this independent of subwords, we can define ω_u^w as

$$\omega_u^w = \sum_{\mathbf{u}, \mathbf{v}} \beta^{|\mathbf{u}| - \ell(u)} \omega_{\mathbf{v}}^{\mathbf{w}/\mathbf{u}}$$

where the sum is over all subwords $\mathbf{u}$ of w whose Hecke product is u , with $\mathbf{v}$ in the complementary spaces of $\mathbf{u}$. Then we have

$$\begin{aligned} \omega^w(fg) &= \sum_u \omega^u(f) \omega_u^w(g) \\ \partial^i \frac{y_i}{1 + \beta y_i} &= \frac{\frac{y_i}{1 + \beta y_i} - \frac{y_{i+1}}{1 + \beta y_{i+1}}}{y_i - y_{i+1}} \\ &= \frac{\frac{y_i}{1 + \beta y_i} - \frac{y_{i+1}}{1 + \beta y_{i+1}}}{y_i - y_{i+1}} \\ &= \frac{1}{(1 + \beta y_i)(1 + \beta y_{i+1})} \\ &+ s_i = \frac{1 + y_{i+1}}{(1 + \beta y_i)(1 + \beta y_{i+1})} \\ &= \frac{1 + \beta y_{i+1} - (\beta - 1)y_{i+1}}{(1 + \beta y_i)(1 + \beta y_{i+1})} \\ &= \frac{1}{1 + \beta y_i} - \frac{(\beta - 1)y_{i+1}}{(1 + \beta y_i)(1 + \beta y_{i+1})} \\ \partial^i \otimes 1 + s_i \otimes \partial^i + s_i \otimes s_i \partial^i \otimes 1 + s_i \otimes (\partial^i + s_i) \\ (\partial^i + s_i) \otimes 1 + s_i \otimes (\partial^i + s_i) - s_i \otimes 1 \end{aligned}$$

$$\pi^i = \frac{(1 + x_{i+1}) - (1 + x_i)s_i}{x_i - x_{i+1}}$$

$$\partial^i(1 + x_{i+1}) = (1 + y_1)\partial^i(x_{i+1} \ominus y_1) + y_1\partial^i = (x_i \ominus y_1)\partial^i + y_1\partial^i - 1$$

$$\mathfrak{S}_\mu(x; y) = y^{-|W|} \mathfrak{S}_\mu(x; y)$$

$$\pi^p \dots \pi^1 = \partial^p \dots \partial^1 E_p(x; 1)$$

Write

$$w = c[\mathfrak{c}_n^*(w); n] \dots c[\mathfrak{c}_1^*(w); 1]$$

$$\mathfrak{S}_\mu(y; 1)^{-1} \partial^{\mathfrak{c}_1^*(w)} \sum E_{p-\ell(\mu, v)}(y_{P(\mu, v)}; 1) \mathfrak{S}_v(x; y)$$

$$\begin{aligned} \pi^i &= \frac{1}{x_i - x_{i+1}} ((1 + x_{i+1}) - (1 + x_i)s_i) \\ &\quad (1 + \beta x_i) \partial^i - \beta \end{aligned}$$

$$\mathfrak{S}_\mu(x; y) = (1 \ominus 0)^W \mathfrak{S}_\mu(x; y)$$

$$\begin{aligned} \mathfrak{S}_\mu(x; y) &= (1 \ominus 0)^W \mathfrak{S}_\mu(x; y) \\ &\quad \partial^i(1 + x_{i+1} - x_i + x_i) \end{aligned}$$

$$(1 + y)^\mu \partial^i (1 + \beta x_{i+1}) \mathfrak{S}_\mu(x; y)$$

Claim: Every Grothendieck polynomial $\mathfrak{S}_w(x)$ can be expressed as a sum of products of ordinary elementary symmetric stable Grothendieck polynomials with integer coefficients, with the number of factors of the product fixed (allowing degree 0 factors) each being in a fixed number of x variables.

Let $\mathfrak{K}^i$ denote the operator $(1 + y_{i+1})\partial^i - 1$, acting on the y variables. Consider the product of operators

$$\mathfrak{K}^{i+p} \dots \mathfrak{K}^{i+1} \mathfrak{K}^i = ((1 + y_{i+p})\partial^{i+p} - 1) \dots ((1 + y_{i+1})\partial^{i+1} - 1)((1 + y_i)\partial^i - 1) E_p(\tau^i y; 1)$$

We have that the top degree factorial elementary stable Grothendieck polynomial in k variables is

$$\mathfrak{E}_k(x; y) = E_k\left(\frac{x}{1+x}; y_1\right) E_k(x; 1)$$

where E_k is the ordinary top degree factorial elementary symmetric polynomial. Thus, for a dominant permutation, applying such a product of operators gives

$$\begin{aligned} & \mathfrak{R}^{i+p} \dots \mathfrak{R}^{i+1} \mathfrak{R}^i \prod_{j=1} E_{\lambda_j}(x; y_j) \\ &= \partial^{i+p} \dots \partial^{i+1} \partial^i E_p(\tau^i y; 1) \prod_{j=1} E_{\lambda_j}\left(\frac{x}{1+x}; y_j\right) E_{\lambda_j}(x; 1) \end{aligned}$$

Applying the Schubert formulas and setting $y_i = 0$ yields a sum of products of ordinary elementary symmetric stable Grothendieck polynomials with the same length as λ with the i th factor having λ_i x variables.

Abbreviate such a strip by $\mathfrak{R}^{[i;p]}$. For each permutation u , we have that

$$\mathfrak{R}^u = \mathfrak{R}^{[1; c_1(u)]} \mathfrak{R}^{[2; c_2(u)]} \dots \mathfrak{R}^{[n; c_n(u)]}$$

Iterating the above argument, we obtain the claim.

$$\begin{aligned} X_i &= X_i - 1 \\ Y_i &= y_i + 1 \end{aligned}$$

$$x_i + y_j + x_i y_j$$

$$h_1^2 - h_2 + h_1$$

$$\begin{aligned} x_i \oplus y_j &= Y_j(X_i - \frac{1}{Y_j}) \\ Z_j^{-1}(X_i - Z_j) \end{aligned}$$

$$\sum_i (1+y_1)^i E_{i,k}(x; y) E_{k-i}(y(1+y_1); -y_1)$$

$$\prod_{i=1}^{n-1} E_{n-i}(x \oplus y_i)$$

$$\mathfrak{S}_{w_0}(x; y) = \sum_{u \in S_n} \partial_{w_0 u^{-1}}^{w_0 w^{-1}} (\mathfrak{S}_{w_0}(x; y)) \mid_{x=y} \mathfrak{S}_u(x; y)$$

8. ABSTRACT PIERI RINGS

[6]

REFERENCES

- [1] BUCH, A. S., KRESCH, A., SHIMOZONO, M., TAMVAKIS, H., AND YONG, A. Stable Grothendieck polynomials and K -theoretic factor sequences. *Mathematische Annalen* 340, 2 (2008), 359–382.
- [2] BUCH, A. S., AND SAMUEL, M. J. K -theory of minuscule varieties. *Journal für die reine und angewandte Mathematik (Crelle's Journal)* 2016, 719 (2016), 133–171. arXiv:1306.5419.
- [3] GAO, Y., AND HUANG, D. The canonical bijection between pipe dreams and bumpless pipe dreams. *International Mathematics Research Notices* 2023, 21 (November 2023), 18629–18663.
- [4] HUANG, D., SHIMOZONO, M., AND YU, T. Marked bumpless pipedreams and compatible pairs. *arXiv preprint arXiv:2407.18160* (2024).
- [5] REINER, V., AND YONG, A. The “Grothendieck to Lascoux” conjecture. *arXiv preprint arXiv:2102.12399* (2021).
- [6] SAMUEL, M. J. A Molev-Sagan type formula for double Schubert polynomials. *J. Pure Appl. Algebra* 228, 7 (2024), 107636.
- [7] SHIMOZONO, M., AND YU, T. Grothendieck to Lascoux expansions. *arXiv preprint arXiv:2106.13922* (2021).
- [8] WEIGANDT, A. Bumpless pipe dreams and alternating sign matrices. *Journal of Combinatorial Theory, Series A* 182 (August 2021), 105470.